

# CSE 4412: Introduction to Quantum Computing

## Homework 1 — Chapter 2 Exercises

### Exercise 2.2

**Statement.** Let  $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$  and  $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ . Compute  $\langle +|+\rangle$ ,  $\langle +|-\rangle$ , and  $\langle -|-\rangle$ . What can you say about these two vectors? You may use the fact that  $|0\rangle$  and  $|1\rangle$  are orthonormal.

**Hint.** Expand each bracket using bilinearity of the inner product (distribute over the  $\pm$  sums), then substitute  $\langle 0|0\rangle = \langle 1|1\rangle = 1$  and  $\langle 0|1\rangle = \langle 1|0\rangle = 0$ . Once you have all three numbers, think about what “normalized” and “orthogonal” mean (Definitions in Section 2.1.2) and what it implies when two normalized, mutually orthogonal vectors span the whole space.

**Your answer:**

Expand each inner product using bilinearity, then substitute  $\langle 0|0\rangle = \langle 1|1\rangle = 1$  and  $\langle 0|1\rangle = \langle 1|0\rangle = 0$ .

$$\langle +|+\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

$$\langle +|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1 - 0 + 0 - 1) = 0$$

$$\langle -|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle - \langle 0|1\rangle - \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 - 0 - 0 + 1) = 1$$

$\langle +|+\rangle = \langle -|-\rangle = 1$  means both  $|+\rangle$  and  $|-\rangle$  are unit vectors, and  $\langle +|-\rangle = 0$  means they’re orthogonal. Two normalized, mutually orthogonal vectors in a 2D space form a basis, so  $\{|+\rangle, |-\rangle\}$  works as an orthonormal basis for the qubit just as well as  $\{|0\rangle, |1\rangle\}$  does.

### Exercise 2.4

**Statement.** Consider the following three questions:

1. Consider the  $|+\rangle$  state. What is the probability of observing  $|1\rangle$  after a measurement?
2. Consider  $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ . What is the probability of observing  $|1\rangle$  after measuring that state?
3. Consider the state  $|1\rangle$ . What is the probability of observing  $|1\rangle$  after measurement?

**Hint.** For a state  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ , the rule from the chapter is  $\Pr(|1\rangle) = |\beta|^2$ , where  $\beta$  is whatever is multiplying  $|1\rangle$  once the state is written in the  $\{|0\rangle, |1\rangle\}$  basis. Identify  $\beta$  in each of the three given states (watch the sign in  $|-\rangle$  — remember you square the *magnitude*).

**Your answer:**

For a state  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ ,  $\Pr(|1\rangle) = |\beta|^2$ .

1.  $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ , so  $\beta = \frac{1}{\sqrt{2}}$  and

$$\Pr(|1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

2.  $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$ , so  $\beta = -\frac{1}{\sqrt{2}}$ . The minus sign disappears once we square the magnitude:

$$\Pr(|1\rangle) = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

Same answer as for  $|+\rangle$ , even though  $|+\rangle$  and  $|-\rangle$  are orthogonal (Exercise 2.2) — flipping the sign between  $\alpha$  and  $\beta$  doesn't show up in computational-basis measurement probabilities.

3.  $|1\rangle = 0 \cdot |0\rangle + 1 \cdot |1\rangle$ , so  $\beta = 1$  and  $\Pr(|1\rangle) = |1|^2 = 1$ : measuring  $|1\rangle$  gives “1” every time, which makes sense since it's already a basis state.

## Exercise 2.5

**Statement.** Imagine you are given a qubit  $|\psi\rangle$  but not told anything about it (i.e., you do not know its probability amplitudes). If you perform a measurement on the state and the outcome is “0,” what does this tell you about the state? Can you learn the original probability amplitudes? What if you are given as many copies of  $|\psi\rangle$  as you want — what can be said then?

**Hint.** Separate the question into two parts. (a) Single measurement: think about *state collapse* — what do you know about the state right after the measurement, versus what you knew about it beforehand? Can one classical bit of outcome data pin down two continuous amplitudes  $\alpha, \beta$ ? (b) Many copies: if you measured each copy and tallied the fraction of “0” vs. “1” outcomes, what would that fraction converge to as the number of copies grows (law of large numbers)? Does that let you recover  $\alpha$  and  $\beta$  exactly, or only something about their magnitudes? Also recall the no-cloning theorem (mentioned in the text, Section 2.5.4) — why can't you manufacture the extra copies yourself?

**Your answer:**

Let  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  with  $\alpha, \beta$  unknown,  $|\alpha|^2 + |\beta|^2 = 1$ .

**One measurement.**  $\alpha$  and  $\beta$  are really two continuous numbers (normalization removes one degree of freedom, and global phase is unobservable, but there's still the relative phase plus a magnitude left to pin down). A single measurement only gives us one classical bit — “0” or “1” — and one bit can't encode two continuous numbers. All we learn from seeing “0” is that  $\alpha \neq 0$ : this outcome had probability  $|\alpha|^2 > 0$ . Worse, the measurement collapses the qubit to exactly

$|0\rangle$ , so whatever information was in the original amplitudes is gone –  $(\alpha, \beta)$  could have been  $(1, 0)$ ,  $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ , or anything else with  $\alpha \neq 0$ , and we'd have seen the same outcome and be left holding the same post-measurement state  $|0\rangle$ .

**Many copies.** Say we get  $n$  independent copies of  $|\psi\rangle$  and measure each one. The fraction that come out "0" has expected value  $|\alpha|^2$ , and by the law of large numbers that fraction converges to  $|\alpha|^2$  as  $n$  grows (similarly the "1" fraction converges to  $|\beta|^2$ ). So with enough copies we can pin down  $|\alpha|$  and  $|\beta|$  as precisely as we like. What we still can't get is the relative phase between  $\alpha$  and  $\beta$ : measuring in the  $\{|0\rangle, |1\rangle\}$  basis can't tell  $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$  apart from  $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ , even though they're orthogonal states (see Exercise 2.4, parts 1 and 2 give the same probability). To catch that phase you'd need copies measured in a different basis too, e.g.  $\{|+\rangle, |-\rangle\}$ . Do this in enough bases and you can reconstruct  $\alpha, \beta$  up to an overall global phase – which is the one thing you can never measure, since  $|\psi\rangle$  and  $e^{i\theta}|\psi\rangle$  are physically identical.

One more thing: the problem says "given" copies for a reason. The no-cloning theorem rules out making those copies yourself – there's no way to take a single unknown  $|\psi\rangle$  and produce two copies of it. So you can't dodge the single-measurement problem by cloning your one qubit before measuring; the extra copies have to come from someone re-preparing the same state independently each time.

## Exercise 2.6

**Statement.** Let  $p \in [0, 1]$  be a real number and consider the state

$$|\psi\rangle = \sqrt{p} |0\rangle + \sqrt{1-p} |1\rangle.$$

Answer the following: (1) probability of observing  $|0\rangle$  for a given  $p$ ; (2) probability of observing  $|-\rangle$  for a given  $p$ ; (3)–(4) graph both probability distributions as functions of  $p$ .

**Hint.** For (1), same rule as Exercise 2.4, just applied to  $|0\rangle$  instead of  $|1\rangle$ . For (2), you can't just read off a coefficient because  $|-\rangle$  isn't one of the two basis vectors  $|\psi\rangle$  is written in terms of — instead use the inner-product rule  $\Pr(|-\rangle) = |\langle -|\psi\rangle|^2$  from Section 2.3.1 (Equation 2.36), and expand  $\langle -|\psi\rangle$  using linearity and the known values of  $\langle 0|0\rangle, \langle 0|1\rangle, \langle 1|0\rangle, \langle 1|1\rangle$ . For (3)–(4), once you have closed-form expressions in terms of  $p$ , you can plot them with pgfplots (`\usepackage{pgfplots}`) or by hand on  $[0, 1]$  — try a few sample points ( $p = 0, \frac{1}{2}, 1$ ) first to sanity-check the shape before you commit to plotting code.

**Your answer:**

(1)  $\Pr(|0\rangle)$ .  $|\psi\rangle = \sqrt{p} |0\rangle + \sqrt{1-p} |1\rangle$ , so  $\alpha = \sqrt{p}$  and

$$\Pr(|0\rangle) = |\alpha|^2 = p.$$

That checks out at the endpoints:  $p = 0$  gives  $|\psi\rangle = |1\rangle$  so  $\Pr(|0\rangle) = 0$ , and  $p = 1$  gives  $|\psi\rangle = |0\rangle$  so  $\Pr(|0\rangle) = 1$ .

(2)  $\Pr(|-\rangle)$ .  $|-\rangle$  isn't one of the two basis states  $|\psi\rangle$  is written in, so we can't just read off a coefficient — we need  $\Pr(|-\rangle) = |\langle -|\psi\rangle|^2$ . Expanding  $\langle -| = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)$  against  $|\psi\rangle$ :

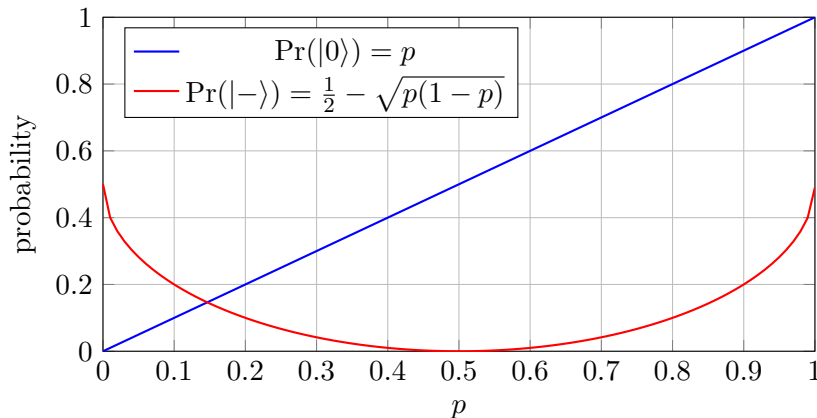
$$\langle -|\psi\rangle = \frac{1}{\sqrt{2}} \left( \sqrt{p} \langle 0|0\rangle + \sqrt{1-p} \langle 0|1\rangle - \sqrt{p} \langle 1|0\rangle - \sqrt{1-p} \langle 1|1\rangle \right) = \frac{1}{\sqrt{2}} (\sqrt{p} - \sqrt{1-p}).$$

Squaring that magnitude:

$$\Pr(|-\rangle) = |\langle -|\psi\rangle|^2 = \frac{1}{2}(\sqrt{p} - \sqrt{1-p})^2 = \frac{1}{2}(p + (1-p) - 2\sqrt{p(1-p)}) = \frac{1}{2} - \sqrt{p(1-p)}.$$

At  $p = \frac{1}{2}$ ,  $|\psi\rangle = |+\rangle$  and this gives  $\Pr(|-\rangle) = 0$ , which makes sense since  $|+\rangle$  and  $|-\rangle$  are orthogonal. At  $p = 0$  or  $1$ ,  $|\psi\rangle$  is  $|0\rangle$  or  $|1\rangle$ , and we get  $\Pr(|-\rangle) = \frac{1}{2}$  – also right, since  $|0\rangle$  and  $|1\rangle$  each overlap  $|-\rangle$  with amplitude  $\frac{1}{\sqrt{2}}$ .

**(3)–(4) Plots.**  $\Pr(|0\rangle) = p$  is just a straight line from  $(0,0)$  to  $(1,1)$ .  $\Pr(|-\rangle) = \frac{1}{2} - \sqrt{p(1-p)}$  is a valley shape: symmetric around  $p = \frac{1}{2}$ , hits 0 right at  $p = \frac{1}{2}$ , and climbs back up to  $\frac{1}{2}$  at both ends.



## Exercise 2.8

**Statement.** Let  $|\psi\rangle$  be some  $d$ -dimensional state and let  $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$  be any orthonormal basis. Show that  $|\psi\rangle$  may be written as

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j|\psi\rangle |b_j\rangle.$$

**Hint.** This is essentially asking you to re-derive the *completeness relation* argument from Section 2.1.3 (Equation 2.22-2.23). Start from the fact that  $\sum_j |b_j\rangle \langle b_j| = I$  for any orthonormal basis, apply both sides to  $|\psi\rangle$ , and note that  $|b_j\rangle \langle b_j||\psi\rangle$  can be regrouped as a scalar  $\langle b_j|\psi\rangle$  times the vector  $|b_j\rangle$ .

**Your answer:**

$\mathcal{B}$  is a basis, so  $|\psi\rangle$  can be written as *some* combination of the  $|b_j\rangle$ :

$$|\psi\rangle = \sum_{j=0}^{d-1} c_j |b_j\rangle$$

for coefficients  $c_j$  we don't know yet. To find them, hit both sides with  $\langle b_k|$  for some fixed  $k$ :

$$\langle b_k|\psi\rangle = \sum_{j=0}^{d-1} c_j \langle b_k|b_j\rangle.$$

Orthonormality means  $\langle b_k | b_j \rangle = \delta_{kj}$ , so every term on the right vanishes except  $j = k$ , leaving  $\langle b_k | \psi \rangle = c_k$ . That holds for every  $k$ , so  $c_j = \langle b_j | \psi \rangle$  for all  $j$ . Plugging back in:

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j | \psi \rangle |b_j\rangle.$$

This is really the same thing as the completeness relation  $\sum_j |b_j\rangle \langle b_j| = I$  – apply it to  $|\psi\rangle$  and regroup each term  $|b_j\rangle \langle b_j| |\psi\rangle$  as  $\langle b_j | \psi \rangle |b_j\rangle$  and you get the identical formula. It's also why  $\langle b_j | \psi \rangle$  gets called the "component of  $|\psi\rangle$  along  $|b_j\rangle$ " – same idea as a dot product picking out coordinates, just for any orthonormal basis, not only  $\{|0\rangle, |1\rangle\}$ .